

Safety Data Sheet

Product name: DL- alpha –Tocopherol

Product code: YYES01、 YYES02、 YYES03、 YYES04、 SPES01

Current version: 2.5, issued: Jun.15,2023 **Replaced version:** 2.4, issued: May.09,2023

Section 1. Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier:

Product name: DL- alpha –Tocopherol
 Product code: YYES01、 YYES02、 YYES03、 YYES04、 SPES01
 Brand: NHU™/COVITAS™
 CAS No.: 10191-41-0
 EC No.: 233-466-0

1.2 Relevant identified uses of the substance or mixture and uses advised against:

Relevant identified uses of the substance or mixture:

Relevant identified uses of the substance or mixture: pharmaceuticals, cosmetics, food supplement.

1.3 Details of the supplier of the safety data sheet:

Company

Zhejiang NHU CO. Ltd.

Address

No.428 Xinchang Dadao West Road, Qixing street,
 Xinchang County, Zhejiang Province, China .312500
 Tel: +86-575-86066811
 E-Mail: sales@cnhu.com
 Website: http://www.cnhu.com/

1.4 Emergency telephone number:

Please contact the regional NHU representation in your country.

Section 2. Hazard identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [EU-GHS/CLP]

Skin sensitization (Category 1), H317

Classification in accordance with Directive 67/548/EEC

None.

Classification according to Hazard Communication Standard (29 CFR 1910.1200)

2.2 Label elements

Labelling according Regulation (EC) No 1272/2008

Pictogram



Signal word

Warning

Hazard statement(s)

H317

May cause an allergic skin reaction.

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Precautionary
statement(s)

P261

Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray.

P272

Contaminated work clothing should not be allowed out of the workplace.

P280

Wear protective gloves.

P302 + P352

IF ON SKIN: Wash with plenty of water.

P333 + P313

If skin irritation or rash occurs: Get medical advice/ attention.

P501

Dispose of contents/container to an approved waste Disposal plant.

Supplemental Hazard
Statements

none

2.3 Other hazards

No data available.

Section 3. Composition and information on ingredients

3.1 Substances

Chemical characterization

Substance name:	DL-alpha-Tocopherol
Synonyms:	Vitamin E, all-rac- α -Tocopherol
Formula:	C ₂₉ H ₅₀ O ₂
Molecular Weight:	430.71 g/mol

Identification numbers

CAS-No. :	10191-41-0
EC-No.:	233-466-0
Index-No.:	Not available

3.2 Mixtures

Not applicable. The product is not a mixture.

Section 4. First aid measures

4.1 Description of first aid measures:

General advice

Show this material safety data sheet to the doctor in attendance.

If inhaled

After inhalation: fresh air.

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In case of skin contact

In case of skin contact: Take off immediately all contaminated clothing. Rinse skin with water/ shower. Consult a physician.

In case of eye contact

After eye contact: rinse out with plenty of water. Remove contact lenses.

If swallowed

If swallowing: immediately make victim drink water (two glasses at most). Consult a physician.

4.2 Most important symptoms and effects, both acute and delayed

The most important known symptoms and effects are described in the labelling (see section 2.2) and/or in section 11.

4.3 Indication of any immediate medical attention and special treatment needed

No data available.

Section 5. Fire-fighting measures

5.1 Extinguishing media:

Suitable extinguishing media

Carbon dioxide (CO₂) Foam Dry powder.

Unsuitable extinguishing media

For this substance/mixture no limitations of extinguishing agents are given.

5.2 Special hazards arising from the substance or mixture:

Carbon oxides

Development of hazardous combustion gases or vapors possible in the event of fire.

Vapors are heavier than air and may spread along floors.

Forms explosive mixtures with air on intense heating.

5.3 Special protective equipment for fire-fighters:

Stay in danger area only with self-contained breathing apparatus. Prevent skin contact by keeping a safe distance or by wearing suitable protective clothing.

5.4 Further information:

Prevent fire extinguishing water from contaminating surface water or the ground water system.

Section 6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures:

Advice for non-emergency personnel: Do not breathe vapors, aerosols. Avoid substance contact.

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Ensure adequate ventilation. Evacuate the danger area, observe emergency procedures, consult an expert.

For personal protection see section 8.

6.2 Environmental precautions:

Do not let product enter drains.

6.3 Methods and material for containment and cleaning up:

Cover drains. Collect, bind, and pump off spills. Observe possible material restrictions (see sections 7 and 10). Take up with liquid-absorbent material (e.g. Chemizorb®). Dispose of properly. Clean up affected area.

6.4 Reference to other sections:

For disposal see section 13.

Section 7. Handling and storage

7.1 Precautions for safe handling:

For precautions see section 2.2.

7.2 Conditions for safe storage, including any incompatibilities:

Suitable: Keep tightly closed.

Store : Preserve in tight containers, protected from light. Protect DL-alpha-Tocopherol with a blanket of an inert gas.

7.3 Specific end use(s):

Air and light sensitive.

Section 8. Exposure controls/Personal protection

8.1 Control parameters:

Ingredients with workplace control parameters.

8.2 Exposure controls

Personal protective equipment

Eyel/face protection

Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU). Safety glasses

Skin protection

Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands.

The selected protective gloves have to satisfy the specifications of Regulation (EU) 2016/425 and the standard EN 374 derived from it.

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Full contact

Material: Nitrile rubber

Minimum layer thickness: 0,11 mm

Break through time: 480 min

Material tested: Dermatril® (KCL 740 / Aldrich Z677272, Size M)

Splash contact

Material: Nitrile rubber

Minimum layer thickness: 0,11 mm

Break through time: 480 min

Material tested: Dermatril® (KCL 740 / Aldrich Z677272, Size M)

data source: KCL GmbH, D-36124 Eichenzell, phone +49 (0)6659 87300, e-mail sales@kcl.de,

test method: EN374

If used in solution, or mixed with other substances, and under conditions which differ from EN 374, contact the supplier of the EC approved gloves. This recommendation is advisory only and must be evaluated by an industrial hygienist and safety officer familiar with the specific situation of anticipated use by our customers. It should not be construed as offering an approval for any specific use scenario.

Body Protection

Protective clothing

Respiratory protection

Required when vapours /aerosols are generated.

Our recommendations on filtering respiratory protection are based on the following standards: DIN EN 143, DIN 14387 and other accompanying standards relating to the used respiratory protection system.

Recommended Filter type: Filter type ABEK

Control of environmental exposure

Do not let product enter drains.

Section 9. Physical and chemical properties

9.1 Information on basic physical and chemical properties:

a) Appearance	Form: viscous oil liquid Colour: clear colorless or yellowish-brown
b) Odour	odorless
c) Odour Threshold	No data available
d) pH	No data available

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e) Melting point/freezing point	Melting point: 2,5 - 3,5 °C
f) Initial boiling point and boiling range	208 - 210 °C at 0,0266 hPa
g) Flash point	>250 °C - closed cup
h) Evaporation rate	No data available
i) Flammability (solid, gas)	No data available
j) Upper/lower flammability or explosive limits	No data available
k) Vapour pressure	13 hPa at 155 °C
l) Vapour density	No data available
f) Initial boiling point and boiling range	0,95 g/mL at 20 °C
n) Water solubility	practically insoluble in water
o) Partition coefficient: noctanol/water	No data available
p) Auto-ignition temperature	No data available
q) Decomposition temperature	No data available
r) Viscosity	No data available
s) Explosive properties	No data available
t) Oxidizing properties	No data available

9.2 Other information:

No data available

Section 10. Stability and reactivity

10.1 Reactivity:

No data available.

10.2 Chemical stability:

Exposure to light.

Stable under recommended storage conditions.

10.3 Possibility of hazardous reactions:

No data available.

10.4 Conditions to avoid:

Exposure to air.

10.5 Incompatible materials:

No data available

10.6 Hazardous decomposition products:

In the event of fire: see section 5

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Section 11. Toxicological information

11.1 Information on toxicological effects

Acute toxicity

LD50 Oral - Rat - male and female - > 4.000 mg/kg (OECD Test Guideline 401)

LD50 Dermal - Rat - male and female - > 3.000 mg/kg (OECD Test Guideline 402)

Skin corrosion/irritation

Skin - Rabbit

Result: slight irritation

(OECD Test Guideline 404)

Serious eye damage/eye irritation

Eyes - Rabbit

Result: No eye irritation

(OECD Test Guideline 405)

Respiratory or skin sensitisation

Local lymph node assay (LLNA) - Mouse

Result: positive

(OECD Test Guideline 429)

Germ cell mutagenicity

Test type: Mutagenicity (mammal cell test): chromosome aberration.

Test system: Chinese hamster lung cells

Metabolic activation: without metabolic activation.

Method: OECD Test Guideline 476

Result: negative

Remarks: (ECHA)

Test type: Ames test

Test system: S. typhimurium

Metabolic activation: with and without metabolic activation

Method: OECD Test Guideline 471

Result: negative

Test type: Mutagenicity (mammal cell test): chromosome aberration.

Test system: Human lymphocytes

Metabolic activation: with and without metabolic activation

Method: OECD Test Guideline 473

Result: negative

Test type: Mutagenicity (mammal cell test): micronucleus

Species: Mouse

Cell type: Bone marrow

Application Route: Oral

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Method: OECD Test Guideline 474

Carcinogenicity

IARC: No ingredient of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.

Specific target organ toxicity - single exposure

No data available

Specific target organ toxicity - repeated exposure

No data available

Aspiration hazard

No data available

11.2 Additional Information

RTECS: GA8746000

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

Other dangerous properties can not be excluded.

Handle in accordance with good industrial hygiene and safety practice.

Section 12. Ecological information

12.1 Ecotoxicity

Toxicity to fish	static test LC50 - Oncorhynchus mykiss (rainbow trout) - > 100 mg/l - 96 h (OECD Test Guideline 203)
Toxicity to daphnia and other aquatic invertebrates	static test EC50 - Daphnia magna (Water flea) - > 23,53 mg/l - 48 h (OECD Test Guideline 202)
Toxicity to algae	static test ErC50 - Pseudokirchneriella subcapitata (green algae) - > 25,8 mg/l - 72 h (OECD Test Guideline 201)
Toxicity to bacteria	static test EC20 - activated sludge - > 927 mg/l - 30 min (DIN/EN ISO 8192)

12.2 Persistence and degradability:

Biodegradability	aerobic - Exposure time 28 d Result: 20 % - Not readily biodegradable. (OECD Test Guideline 301F)
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12.3 Bioaccumulative potential:

No data available

12.4 Mobility in soil:

No data available

12.5 Results of PBT and vPvB assessment:

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This substance contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Other adverse effects:

Discharge into the environment must be avoided.

Section 13. Disposal considerations

13.1 Waste treatment methods:

Product

Offer surplus and non-recyclable solutions to a licensed disposal company.

Contaminated packaging

Dispose of as unused product.

Section 14. Transport information

14.1 UN number					
ADR/RID:	-	IMDG:	-	IATA:	-
14.2 UN proper shipping name					
ADR/RID:	Not dangerous goods				
IMDG:	Not dangerous goods				
IATA:	Not dangerous goods				
14.3 Transport hazard class(es)					
ADR/RID:	-	IMDG:	-	IATA:	-
14.4 Packaging group					
ADR/RID:	-	IMDG:	-	IATA:	-
14.5 Environmental hazards					
ADR/RID: no		IMDG Marine pollutant: no		IATA: no	
14.6 Special precautions for user					
No data available					

Further information

Not classified as dangerous in the meaning of transport regulations.

Section 15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

This material safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.

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National legislation

Seveso III: Directive 2012/18/EU of the European Parliament and of the Council on the control of major-accident hazards involving dangerous substances

Not applicable

Other regulations

Take note of Dir 94/33/EC on the protection of young people at work.

15.2 Chemical safety assessment

For this product a chemical safety assessment was not carried out.

Section 16. Other information

16.1 Full text of the R-, H- and EUH-phrases drawn up in sections 2 and 3 (provided not already drawn up in these sections):

Acute Tox.

Acute toxicity

H317

May cause an allergic skin reaction.

16.2 Further information

The above information is believed to be correct but does not purport to be all inclusive and shall be used only as a guide. The information in this document is based on the present state of our knowledge and is applicable to the product with regard to appropriate safety precautions. It does not represent any guarantee of the properties of the product. NHU Corporation and its Affiliates shall not be held liable for any damage resulting from handling or from contact with the above product.